

BASINS workflow

1. Environmental observations

Chemistry + CTD profiles

Match profile keys and nearest depth; reconcile oxygen measurements.



2. Physical water-column structure

Density, stratification, and renewal

GSW density; N², mixed-layer depth, PEA, oxygen/nitrate event diagnostics.



3. Environmental feature matrix

Cleaning, missingness, and PCA

Select numeric chemistry; optional missingness sensitivity; scores and loadings.



4. Environmental compartments

GMM, oxygen, and hybrid schemes

Select K; fit responsibilities; combine assignments; compare and subdivide states.



5. Temporal and ecological context

Transitions, succession, and feature associations

Track environmental states; quantify associated measurements and temporal changes.



6. Cruise structure and refinement

EOF groups and within-GMM HDBSCAN

Baseline-adjusted profiles; stability and season checks; optional continuous sections.



7. Optional metabolic modeling

Environmental media + genome reconstruction

gapseq fixed models; COBRApy growth, exchange FVA, nutrient counterfactuals.



8. Published results

Tables, figures, provenance, and HTML report

Nextflow completion → atomic publication into modules, summary, and logs.